

A PROJECT TITLE
ON

Portfolio Risk-Factor Analysis Using PCA for Dimensionality Reduction

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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Name of Title: FINANCE

1. Abstract:

Portfolio risk analysis is an important part of finance that helps investors understand and manage the uncertainty associated with investments. Modern financial markets generate a large amount of data from stock prices, returns, interest rates, exchange rates, and other economic indicators. Analyzing all these variables together can be difficult because many of them are highly correlated. This increases the complexity of financial models and may reduce prediction accuracy. To solve this problem, **Principal Component Analysis (PCA)** is used as an effective dimensionality reduction technique.

This project focuses on applying PCA to identify the major risk factors that influence a financial portfolio. PCA transforms a large set of correlated variables into a smaller number of uncorrelated variables called **principal components**. These components retain most of the important information present in the original dataset while reducing noise and redundancy. By selecting only the most significant components, investors can better understand the key sources of portfolio risk without analyzing every individual variable.

The proposed approach involves collecting historical financial data, preprocessing the dataset by handling missing values and standardizing features, applying PCA, and evaluating the amount of variance explained by each principal component. The reduced dataset can then be used for portfolio risk assessment, visualization, and further financial analysis. This improves computational efficiency and simplifies decision-making while preserving the essential characteristics of the data.

The project demonstrates how PCA helps identify hidden patterns in financial datasets and supports better investment strategies through effective risk-factor analysis. It enables financial analysts and portfolio managers to reduce data complexity, improve model performance, and make informed investment decisions. Overall, this study highlights the importance of dimensionality reduction in finance and shows that PCA is a powerful tool for extracting meaningful information from high-dimensional financial data while maintaining the accuracy and reliability of portfolio risk analysis.

2. INTRODUCTION:-

Portfolio Risk-Factor Analysis is an important area in finance that helps investors identify and manage the risks associated with investment portfolios. Financial markets generate large volumes of data, including stock prices, returns, interest rates, exchange rates, and economic indicators, making analysis complex due to the high correlation among variables. **Principal Component Analysis (PCA)** is a statistical and machine learning technique used for dimensionality reduction, which transforms a large set of correlated variables into a smaller set of uncorrelated variables called principal components while preserving most of the important information. By reducing data complexity, PCA helps identify the major factors influencing portfolio risk, improves computational efficiency, minimizes redundancy, and simplifies financial analysis. This project applies PCA to historical financial data to extract the most significant risk factors, enabling better portfolio evaluation, risk assessment, and investment decision-making. Overall, the use of PCA provides a simple, efficient, and reliable approach for understanding portfolio risk and enhancing financial analysis in modern investment management.

2. Software Requirements:-

Project: Portfolio Risk-Factor Analysis Using PCA for Dimensionality Reduction

1. Operating System

Windows 10/11 (64-bit), Ubuntu 20.04+, macOS Monterey+

2. Programming Language

Python 3.10 or above

3. Development Environment

Visual Studio Code, PyCharm Community Edition, Jupyter Notebook, Google Colab

4. Python Libraries

- NumPy – Numerical computations
- Pandas – Data loading and preprocessing
- Scikit-learn – PCA and ML algorithms
- Matplotlib – Graph plotting

- Seaborn – Statistical visualization
- Plotly – Interactive charts
- SciPy – Mathematical functions
- yFinance – Download stock market data
- OpenPyXL – Excel file support

5. Installation Commands

`pip install numpy pandas scikit-learn matplotlib seaborn scipy yfinance plotly openpyxl`

6. Dataset Requirements

Historical stock prices, returns, interest rates, exchange rates, inflation data, market indices, and sector-wise stock data.

Sources: Yahoo Finance, Kaggle, NSE India, Alpha Vantage API.

7. Database (Optional)

SQLite, MySQL, or PostgreSQL.

8. Data Preprocessing

- Handle missing values
- Remove duplicates
- Standardize features using StandardScaler
- Normalize numerical data
- Convert categorical values if needed

9. Machine Learning Technique

Principal Component Analysis (PCA)

Purpose:

- Reduce dimensionality
- Remove redundant features
- Identify key portfolio risk factors
- Improve computational efficiency

10. Visualization

Scree Plot, Explained Variance Plot, Correlation Heatmap, PCA Scatter Plot, Portfolio Risk Charts using Matplotlib/Seaborn/Plotly.

11. Hardware Requirements

Minimum:

- Intel Core i3 / Ryzen 3
- 4 GB RAM
- 20 GB storage

Recommended:

- Intel Core i5 / Ryzen 5+
- 8 GB RAM
- SSD 256 GB+

12. Version Control

Git and GitHub

13. Web Browser

Google Chrome, Microsoft Edge, Mozilla Firefox

14. Workflow

Financial Data → Data Cleaning → Feature Standardization → PCA → Risk Factor Extraction → Visualization → Portfolio Risk Report

15. Expected Output

- Reduced-dimensional dataset
- Principal Components
- Explained Variance Ratio
- Portfolio Risk Factors
- Risk Visualizations
- Investment Decision Support



Signature of the Guide



Course Instructor